



Maritime &
Coastguard
Agency

Statutory guidance

MSN 1903 (M) GMDSS ship requirements

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1. Introductions and definitions

1.1 The Merchant Shipping (Radiocommunications) (Amendment) Regulations 2021 give effect in UK law to the outstanding amendments to Chapter IV of the Annex to the International Convention for the Safety of Life at Sea, 1974 (“SOLAS”). The ambulatory reference provision in regulation 7 of those Regulations provides for UK law to be amended in line with changes to Convention provisions without the need for further legislation.

1.2 The 1998 Regulations remain in force but no longer apply to ships to which Chapter IV of SOLAS applies. Therefore, this Notice relates only to ships to which Chapter IV does not apply, such as cargo ships of not less than 300 gross tonnage on domestic voyages and passenger ships of classes VI and VI(A) made from glass reinforced plastics or wood.

1.3 In this Notice:

- 1.3.1 Unless a term is defined in paragraph 1.4, it has the same meaning as that set out in regulations 2 and 7 of the 1998 Regulations;
- 1.3. 2 A reference to a numbered paragraph is, unless otherwise stated, a reference to the relevant paragraph in this Notice.

1.4 In this Notice:

“AIS” means automatic identification system.

2. Functional Requirements

2.1 Every ship, while at sea, must be capable of –

- 2.1.1 transmitting ship-to-shore distress alerts by at least two separate and independent means, each using a different radio communication service;

- 2.1.2 receiving shore-to-ship distress alerts;
- 2.1.3 transmitting and receiving ship-to-ship distress alerts;
- 2.1.4 transmitting and receiving search and rescue co-ordinating communications;
- 2.1.5 transmitting and receiving on-scene communications;
- 2.1.6 transmitting and, as required by Regulation 5(1) of the Merchant Shipping (Safety of Navigation) Regulations 2020 (SI 2020/673), receiving signals for locating;
- 2.1.7 transmitting and receiving maritime safety information;
- 2.1.8 transmitting and receiving general radiocommunications to and from shore-based radio systems or networks;
- 2.1.9 transmitting and receiving bridge-to-bridge communications

3. Radio Equipment to be provided for all sea areas

3.1 Every ship must be provided with –

- 3.1.1 a VHF radio installation capable of transmitting and receiving –
 - 3.1.1.1 DSC on the frequency 156.525 MHz (channel 70). Means must be provided to initiate the transmission of distress alerts on channel 70 from the position from which the ship is normally navigated; and
 - 3.1.1.2 radiotelephony on the frequencies 156.300 MHz (channel 6), 156.650 MHz (channel 13) and 156.800 MHz (channel 16);
- 3.1.2 a VHF radio installation capable of maintaining a continuous DSC watch on channel 70 which may be separate from, or combined with, that required by sub-paragraph 6.1.1.1 of this paragraph;
- 3.1.3 a search and rescue locating device capable of operating in the 9 GHz band or on frequencies dedicated for AIS, which –
 - 3.1.3.1 must be so stowed that it can be easily utilised; and
 - 3.1.3.2 are required for a survival craft in accordance with the Merchant Shipping (Life-Saving Appliances and Arrangements) Regulations 2020 (SI 2020/501);

- 3.1.4 if the ship is at sea in any area in which an international NAVTEX service is provided, a receiver capable of receiving international NAVTEX service broadcasts;
- 3.1.5 a radio facility for reception of maritime safety information by a recognised mobile satellite service enhanced group calling system, if the ship is at sea in any area of a recognised mobile satellite service coverage but in which an international NAVTEX service is not provided.
- 3.1.6 A satellite EPIRB provided pursuant to this Notice must be –
 - 3.1.6.1 capable of transmitting a distress alert through the Cospas-Sarsat satellite distress alerting service on 406 MHz band;
 - 3.1.6.2 installed in an easily accessible position;
 - 3.1.6.3 ready to be manually released and capable of being carried by one person into a survival craft;
 - 3.1.6.4 capable of floating free if the ship sinks;
 - 3.1.6.5 capable of being activated manually; and
 - 3.1.6.6 automatically activated when afloat.

3.2 Every passenger ship to which this paragraph applies must be provided with means for two-way on-scene radio communications for search and rescue purposes capable of operating solely on the aeronautical frequencies 121.5 MHz and 123.1 MHz from the position from which the ship is normally navigated.

4. Additional radio equipment to be provided for sea area A1

4.1 In addition to meeting the requirements of paragraph 3, every ship on voyages exclusively in sea area A1 must be provided with a radio installation capable of initiating the transmission of ship-to-shore distress alerts by operation from the position from which the ship is normally navigated, operating either -

- 4.1.1 on VHF using DSC; or
- 4.1.2 a satellite EPIRB, using the Cospas-Sarsat satellite distress alerting service on 406 MHz; as required by paragraph 3.1.6 if it is installed close to, or capable of remote activation from, the position from which the ship is normally navigated; or

- 4.1.3 if the ship is at sea within coverage of MF coast stations equipped with DSC, on MF using DSC; or
- 4.1.4 on HF using DSC; or
- 4.1.5 through a recognised mobile satellite service; this requirement may be fulfilled by:
 - 4.1.5.1 a ship earth station; or
 - 4.1.5.2 the satellite EPIRB, required by paragraph 3.1.6, either by installing the satellite EPIRB close to, or by remote activation from, the position from which the ship is normally navigated.

4.2 The VHF radio installation required by paragraph 3.1.1 must also be capable of transmitting and receiving general radiocommunications using radiotelephony.

5. Additional radio equipment to be provided for sea area A2

5.1 In addition to meeting the requirements of paragraph 3, every ship engaged on voyages beyond sea area A1 but remaining within sea area A2, must be provided with –

- 5.1.1 an MF radio installation capable of transmitting and receiving, for distress and safety purposes, on the frequencies –
 - 5.1.1.1 2,187.5 kHz using DSC; and
 - 5.1.1.2 2,182 kHz using radiotelephony;
- 5.1.2 a radio installation capable of maintaining a continuous DSC watch on the frequency 2,187.5 kHz; such installation may be separate from, or combined with, that required by sub-paragraph 5.1.1.1; and
- 5.1.3 means of initiating the transmission of ship-to-shore distress alerts by a radio service other than MF operating either –
 - 5.1.3.1 a satellite EPIRB using the Cospas-Sarsat satellite distress alerting service on 406 MHz; required by paragraph 3.1.6 if it is installed close to, or capable of remote activation from, the position from which the ship is normally navigated; or
 - 5.1.3.2 on HF using DSC; or
 - 5.1.3.3 through a recognised mobile satellite service by a ship earth station.

5.2 Means must be provided to initiate transmission of distress alerts by the radio installations specified in sub-paragraphs 5.1.1 and 5.1.3 from the position from which the ship is normally navigated.

5.3 The ship must, in addition, be capable of transmitting and receiving general radiocommunications using radiotelephony or direct-printing telegraphy by either –

- 5.3.1 a radio installation operating on working frequencies in the bands between 1,605 kHz and 4,000 kHz or between 4,000 kHz and 27,500 kHz; this requirement may be fulfilled by the addition of this capability to the equipment required by sub-paragraph 5.1.1; or
- 5.3.2 a recognised mobile satellite service ship earth station.

6. Additional radio equipment to be provided for sea area A3

6.1 In addition to meeting the requirements of paragraph 3, every ship engaged on voyages beyond sea areas A1 and A2, but remaining within sea area A3, must be provided with either the following equipment –

Alternative A

- 6.1.1 a recognised mobile satellite service ship earth station capable of –
 - 6.1.1.1 transmitting and receiving distress and safety communications using direct printing telegraphy;
 - 6.1.1.2 initiating and receiving distress priority calls;
 - 6.1.1.3 maintaining watch for shore-to-ship distress alerts, including those directed to specifically defined geographical areas;
 - 6.1.1.4 transmitting and receiving general radiocommunications, using either radiotelephony or direct-printing telegraphy; and
- 6.1.2 an MF radio installation capable of transmitting and receiving, for distress and safety purposes, on the frequencies –
 - 6.1.2.1 2,187.5 kHz using DSC; and
 - 6.1.2.2 2,182 kHz using radiotelephony; and

- 6.1.3 a radio installation capable of maintaining a continuous DSC watch on the frequency 2,187.5 kHz which may be separate from or combined with that required by sub-paragraph 6.1.2.1; and
- 6.1.4 means of initiating the transmission of ship-to-shore distress alerts by a radio service operating either –
 - 6.1.4.1 a satellite EPIRB using the Cospas-Sarsat satellite distress alerting service on 406 MHz; required by paragraph 3.1.6. if it is installed close to, or capable of remote activation from, the position from which the ship is normally navigated; or
 - 6.1.4.2 on HF using DSC; or
 - 6.1.4.3 through a recognised mobile satellite service by an additional ship earth station. or the following equipment -

Alternative B

- 6.1.5 an MF/HF radio installation capable of transmitting and receiving, for distress and safety purposes, on all distress and safety frequencies in the bands between 1,605 kHz and 4,000 kHz and between 4,000 kHz and 27,500 kHz –
 - 6.1.5.1 using DSC;
 - 6.1.5.2 using radiotelephony; and
 - 6.1.5.3 using direct-printing telegraphy; and
- 6.1.6 equipment capable of maintaining DSC watch on 2,187.5 kHz, 8,414.5 kHz and on at least one of the distress and safety DSC frequencies 4,207.5 kHz, 6312 kHz, 12,577 kHz or 16,804.5 kHz; the equipment must be such that it must be possible at any time to select any of these DSC distress and safety frequencies. This equipment may be separate from, or combined with, the equipment required by sub-paragraph 6.1.5 above;
- 6.1.7 means of initiating the transmission of ship-to-shore distress alerts by a radiocommunication service other than HF operating either –
 - 6.1.7.1 a satellite EPIRB through using the Cospas-Sarsat satellite distress alerting service on 406 MHz; this requirement may be fulfilled by the satellite EPIRB, required by paragraph 3.1.6 if it is installed close to, or capable of remote activation from, the position from which the ship is normally navigated; or
 - 6.1.7.2 through a recognised mobile satellite service by a ship earth station; and

- 6.1.8 in addition, means of transmitting and receiving general radiocommunications using radiotelephony or direct-printing telegraphy must be provided by an MF/HF radio installation operating on working frequencies in the bands between 1,605 kHz and 4,000 kHz and between 4,000 and 27,500 kHz; this requirement may be fulfilled by the addition of this capability in the equipment required by sub-paragraph 6.1.5.

6.2 Means must be provided to initiate transmissions of distress alerts from the position from which the ship is normally navigated by the radio installations specified in sub-paragraphs 6.1.1, 6.1.2 and 6.1.4 of ALTERNATIVE A and 6.1.5 and 6.1.7 of ALTERNATIVE B of this paragraph.

7. Additional radio equipment to be provided for sea area A4

7.1 In addition to meeting the requirements of paragraph 3, ships engaged on voyages in all sea areas must be provided with the radio installations and equipment specified in ALTERNATIVE B in paragraph 6.1, except that the equipment required by sub-paragraph 6.1.7.2 of ALTERNATIVE B must not be accepted as an alternative to that required by sub-paragraph 6.1.7.1 of ALTERNATIVE B, which must always be provided. Such ships must in addition comply with the requirements of paragraph 6.2.

More Information

UK Technical Services Navigation
Maritime and Coastguard Agency
Bay 2/25
Spring Place
105 Commercial Road
Southampton
SO15 1EG

Telephone; +44 (0) 203 8172000

Email: navigationsafety@mcga.gov.uk



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